A Bayesian DAG is a map I can learn from

Professor, I am using a Bayesian DAG to connect information about a student with their final course result. I find it helpful to think of this as two parts: a map, and a way to learn from data.

The DAG is the map. Each box represents something I measure, and the arrows show the relationships I assume. DAG stands for directed acyclic graph. In plain language, the arrows have a direction, and following them never brings me back to where I started.

The picture shows one path in my larger model: wellness connects to confidence, confidence connects to online activity, activity connects to assessment scores, and those scores connect to the final result. I chose these connections; the arrows do not prove that one thing causes another.

The Bayesian part describes how I learn the strengths of these relationships. I start with cautious estimates and update them using earlier students’ data.

With that idea in place, I will show you what information I put into the map.

I fill the map with four kinds of information

To fill the map, I brought together the grade records, student surveys and course website activity. Although the files contain thousands of repeated entries, my final modeling table has 191 students across six terms, from Fall 2024 through Spring 2026.

I grouped the inputs into four areas. Wellness describes how students reported feeling. Motivation includes their confidence in learning, the value they see in the course, and their ratings of barriers and support. Online activity includes page views and recorded participation. Assessment history includes quiz, paper and other assignment scores.

I converted assignment marks into percentages so scores with different maximum points could be compared, and I removed repeated activity records. I then compared a smaller set of ten assessment measures with a larger set of sixteen.

The surveys and activity summaries use the first four assumed teaching weeks. However, I still need to confirm when each assessment score was released. That means this is a study using past records, rather than a verified week-four forecast.

Before I explain the learning step, I will show how these inputs are connected in the original graph.

The original DAG used in my notebook

Professor, this is the structure used in my notebook. The first slide showed one path to introduce the idea. This slide brings the other connections back into view.

On the left are six measured inputs. Wellness connects to the support and barrier score, and to confidence. The support and barrier score also connects to confidence and page views. Confidence connects to the value students see in the course and to page views. Course value also connects to page views, and page views connect to recorded online participation.

The notebook calls confidence expectancy. It calls the support and barrier score cost_support; higher answers generally describe fewer barriers or better support. The two activity measures are transformed page views and participation on the course website. They do not directly measure MLC use.

On the right, I grouped sixteen assessment variables so the slide remains readable. The model still keeps sixteen separate scores. Confidence and page views both connect to every one of them, giving thirty-two arrows. Each assessment score then connects to the final course percentage. All six inputs on the left also connect directly to that final result.

Together, that gives twenty-three variables and sixty-two arrows. The arrows say which relationships I allow the model to use. They do not show how strong those relationships are, or prove cause and effect. I chose the structure; the data estimate the strengths.

This also explains why including wellness in the graph does not automatically mean it improves predictions. That needs the grades-only comparison we discuss later. Next, I will show how I learn the relationships and produce a forecast.

I learn from past students, then make a forecast

I keep the arrows fixed. For each box, the model learns how much weight to give the inputs pointing into it, and how much variation those inputs do not explain.

I start with cautious estimates and update them using earlier students’ inputs and known final results. To predict another student’s result, I use their available inputs while leaving their final result unknown.

The output is a predicted percentage and a range. In this historical example, the estimate was 83.1 percent and the reported range was about 77.2 to 89.0. The student’s final result was 81.0 percent, which I show here only to check the forecast afterward.

A 90 percent range is meant to include the actual result for about nine out of ten similar students. I have to check whether that happens; it is not a guarantee for one student. I adjusted these ranges using earlier prediction errors, but they still depend on simplified assumptions.

Missing surveys also need care. I used typical training values when learning, while allowing unknown inputs to remain uncertain when predicting.

Before I show how I checked the forecast, I will connect these steps to the technical methods used in the code.

Technical methods behind my Bayesian DAG

Professor, this slide connects the technical names to the steps in my model. I chose the arrows in advance. The data learn their strengths rather than discovering a new graph.

For each box, I use a linear Gaussian model. This means a weighted combination of its inputs, plus bell-shaped variation that those inputs do not explain. This keeps the marks as continuous percentages.

Normal–Inverse-Gamma shrinkage gives me Bayesian estimates of both the weights and that unexplained variation. It discourages large weights unless the data support them. That matters because the dataset is small and many assignment marks carry similar information. I compared coefficient precisions of one and ten; the earlier-term checks selected ten.

For prediction, I condition the fitted joint Gaussian on a student’s known inputs. This calculates their expected final percentage and its remaining uncertainty. If a prediction input is missing, I marginalize it, meaning I average over its possible values within the model. During training, however, missing inputs were filled with training medians; that is still an approximation.

I return a mean and a 90 percent range. The basic Gaussian range is the mean plus or minus 1.645 standard deviations. I also checked rescaling it using earlier prediction errors, which produced the range in my example. The reported values are limited to zero through one hundred.

These main ranges use fitted parameter estimates. They do not include all parameter uncertainty. The notebook adds a separate Student-t check for complete inputs, but not a full missing-data solution.

With these methods explained, I can show how I checked the predictions.

I learn from earlier terms and check later students

To check whether this approach can work beyond the students it learned from, I followed the order of the course terms. Within the older terms, I repeatedly learned from earlier classes and checked the next class. That helped me choose how many assessment scores to include and how cautious the model should be.

After making those choices, I used the selected model to learn from 148 students from the first five terms. I then predicted the final percentages of 43 students from Spring 2026, without using their final results to fit that model.

The timeline shows that separation. I also kept the data preparation within the training group, so the later students did not determine the values used to prepare the earlier data.

Spring 2026 was kept out of fitting, but its results have already been reviewed during this project. That does not mean I trained on those students. It means I should be careful about treating later changes to the approach as independently confirmed by that same class.

I will work with the six terms already available. They allow four checks where I learn from earlier terms and predict the next term, starting with Spring 2025. I will report how the results vary between those checks.

With that testing process explained, I can now show the results and what they do and do not tell me.

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The extra inputs have not improved average error yet

The numbers in this table measure prediction mistakes. They are not the students' quiz marks or final grades. I calculate each mistake by taking the distance between a predicted percentage and the actual final percentage, then average those distances across the 43 students.

The first approach gives everyone the same forecast: 79.92 percent, which is the average final result of the 148 earlier students. Its average mistake is 8.13 percentage points.

The second approach uses each student's own assessment results, so different students receive different forecasts. Its average mistake is 2.82 points. For example, one student finished with 95.34 percent. The assessment-based forecast was about 94.00 percent, whereas the first approach still predicted 79.92.

The third approach also includes wellness, motivation and online activity. Its average mistake is 2.83 points, so those extra inputs have not improved this measure here. Assessment scores are already part of the final grade, which helps explain why they carry so much information.

The reported ranges included the actual result for 39 of the 43 students. However, the largest individual mistake was about 25.7 points, so the average does not describe every student equally well.

This leads to my next question: could the extra inputs help earlier, when fewer assessment results are available?

Next, I will test the extra inputs earlier

My next experiment will focus on a time when the information could still be useful for student support. I would first choose a prediction date, such as the end of week four, and verify which surveys, activity records and assessment scores were actually available by that date.

I would then predict the same students using the same available grades. I would add wellness alone, motivation alone, and online activity alone, and finally compare a model that includes all of them. I would keep the rest of the comparison consistent so an improvement cannot simply come from using later grades or a different set of students.

I would repeat this process across the later terms that are already in the dataset and check both the average error and the prediction ranges. I would also investigate the 78 students who are missing early survey information, because that may affect the comparison.

For each check, I would choose the model settings using earlier terms only, then predict the next term. Since this project uses existing data that have already been explored, I will describe the findings as evidence from past classes. If the additional inputs do not help, that is still a valid result.

If score release dates cannot be recovered, I can still compare clearly named sets of scores, while stating that I cannot confirm how early those predictions could have been made.

I would like your feedback on the prediction date and what evidence would make this useful for early student support.